1. Executive Summary

This report describes the development and evaluation of deep learning models for breast ultrasound (BUS) lesion segmentation, intended for integration with a robotic ultrasound probe simulation. Two architectures were investigated: a custom UNet++ with ResNet34 encoder and nnU-Net v2. Both models were trained on the BUS-BRA dataset (1,875 images) and externally validated on the BrEaST dataset (252 evaluable images). Domain shift remains the primary challenge

UNet++

- UNet++ achieved 87.40 ± 0.85 internal Dice and 70.05 ± 29.28 external Dice
- Inference time: ~25 ms forward pass on Colab (pure inference, not cross-validation), ~106 ms with TTA and post-processing

nnU-Net

- nnU-Net (1000 epochs, 5 fold) achieved 91.14% internal Dice and 76.76% external Dice (**Standardized train+test resolution (320×320)**)
- nnU-Net inference time will be added soon.

2. Dataset Details

BREAST: https://www.cancerimagingarchive.net/collection/breast-lesions-usg/
Article: https://pubmed.ncbi.nlm.nih.gov/38297002/

Bus-Bra:https://www.kaggle.com/datasets/orvile/bus-bra-a-breast-ultrasound-dataset
https://zenodo.org/records/8231412
Article: https://pubmed.ncbi.nlm.nih.gov/37937827/

FEATURE	BUS-BRA	Breast-Lesions-USG
Role	Training and 5-fold CV	External test only
Images	1875	256 (252 with lesion masks used for evaluation, 4 normal (images without any tumors) images excluded)
Masks	One binary mask per image	One or more masks per image (tumor and other lesions), merged into one binary mask

Mean size (H x W)	391 x 321 (min 268 x 240, max 478 x 579)	523 x 620 (min 289 x 363, max 618 x 1024)
Lesion area per image	8.6%	7.2%
Patient number	1064	256
Distrubution of dataset	722 benign and 342 malignant	154 benign tumors, 98 malignancies and 4 normal
BI-RADS	BI-RADS categories (2, 3, 4, and 5)	BI-RADS 1,2,3,4,5
Reliability	biopsy-proven, ground truth segmentation	biopsy-proven
Code availability	https://github.com/wgomezf/BUS-BRA (DeepLabV3+ model with ResNet18 and ResNet50 back-bone)	https://github.com/best-ippt-pan-pl/BrEaST
Segmentation mask	Binary	Binary (Tumor + masks for other lesions)
Probe type	linear array transducer	linear array transducer
Literature	CAD, Deep Learning	BI-RADS analysis, multiple lesion detection

3) Model Details

UNet++

Environment: Colab/T4

Features: UNet++ (ResNet34)

- **Preprocessing:** grayscale PNG, resized to 256x256 (cv2.resize, bilinear; aspect ratio not preserved), intensities divided by 255 (no per-image min-max). Masks are resized the same way and re-binarized at 0.5. BrEaST masks of one image are merged into one binary mask.
- **Architecture:** UNet++ from segmentation_models_pytorch with a ResNet34 encoder, ImageNet-pretrained weights (first convolution adapted to 1 channel), 1 input channel, 1 output channel, sigmoid output.
- **Loss:** Dice + Focal (alpha 0.5, gamma 2). The sigmoid is applied twice (model output and loss); kept as in the experiments.
- **Optimization:** Adam, learning rate 1e-4, weight decay 1e-4, ReduceLROnPlateau on validation loss (factor 0.5, patience 3), batch size 8, at most 50 epochs, early stopping with patience 5 on validation Dice. The best epoch by validation Dice is saved.
- **Cross-validation:** 5-fold, patient-wise (KFold over BUS-BRA patient IDs, shuffle, random_state 42; the patient order comes from a Python set, so fold composition is not reproducible between runs).
- **Training augmentation:**

Transform	Probability	Parameters
Rotation by 90 degree multiples	0.5	k = 1, 2 or 3
Horizontal flip	0.5	
Vertical flip	0.7	

Speckle noise (multiplicative)	**0.7**	**std ~ U(0.02, 0.1)**
Gamma correction	**0.5**	**gamma ~ U(0.7, 1.5)**
CLAHE	**0.4**	**clip limit ~ U(1.5, 3.0), 8x8 grid**
Brightness scaling	**0.5**	**factor ~ U(0.7, 1.3)**
Gaussian blur	**0.3**	**kernel 3 or 5**

- **Inference:** fold 3 checkpoint (best internal Dice), test-time augmentation (original, horizontal flip, vertical flip, 90-degree rotation; probabilities averaged), threshold 0.5, post-processing (connected components smaller than 100 pixels removed, then 3x3 morphological closing).
- **Evaluation:** per-image Dice at 256x256 on the 252 BrEaST images with lesion masks.
- **Speed (Google Colab, Tesla T4, 256x256, batch size 1, excludes image loading, n = 242 after 10 warm-up images):** single forward pass 24.8 ± 3.1 ms/image; TTA (4 passes) + post-processing 105.5 ± 4.9 ms/image.
- **Hardware:** UNet++ was trained and evaluated on Google Colab (Tesla T4).

UNet++	**Dice**
BUS-BRA 5-fold CV (best-epoch validation Dice)	**87.40 ± 0.85 (folds: 88.00, 85.75, 88.04, 87.77, 87.44)**
BrEaST external, mean ± std over images	**70.05 ± 29.28**
BrEaST external, median	**83.45**
Images with Dice ≥ 0.8 / < 0.5	**150 (59.5%) / 54 (21.4%)**

nnU-Net v2

Performed on the workstation GPU

The preprocessing/resampling strategy for both datasets standardized, a consistent image-size or pixel-scale normalization protocol defined before nnU-Net training and then the model retrained.

Step 1: Create environment

Step 2: Create folders (nnUNet_raw, nnUNet_preprocessed, nnUNet_results)

Step 3: Prepare Data (Resize both train and test sets/ NIfTI conversion)

Step 4:Preprocessing

Step 5: Training (5-fold)

Step 6 : Prediction

Step 7: Evaluate

Features:

Architecture: nnU-Net v2 2d configuration with default plans (not ResEnc): PlainConvUNet, 7 stages, features per stage 32-64-128-256-512-512-512, 3x3 convolutions, 2 conv layers per stage, strides 1-2-2-2-2-2-2, InstanceNorm, LeakyReLU, no dropout. Patch size 320x320, batch size 32, batch Dice.

Preprocessing:

- grayscale PNG, min-max normalized to 0-255,
- resized to 320x320 for both datasets (bilinear for images, nearest-neighbor for masks, then re-binarized), saved as NIfTI (H, W, 1) with identity affine
- The aspect ratio is not preserved and no physical pixel spacing exists, so nnU-Net sees spacing 1.0 x 1.0. nnU-Net then applies per-image z-score normalization.

Training:

- 5-fold cross-validation,
- 1000 epochs per fold,

- default nnUNetTrainer (nnU-Net v2 defaults, unchanged: SGD with Nesterov momentum 0.99, initial learning rate 0.01 with polynomial decay, weight decay 3e-5, Dice + cross-entropy loss, deep supervision, default data augmentation).

Inference: ensemble of the 5 fold models with default mirroring test-time augmentation. Post-processing (largest connected component) did not improve cross-validation results and was not used.

Evaluation: per-image Dice, IoU, precision, recall, specificity and accuracy on binarized masks at 320x320, averaged over the 252 external images. Internal Dice (91.14%) is the mean cross-validation Dice reported by nnU-Net.

Notes: (1) Scale consistency matters: external Dice was 66.4% with native image sizes, 74.8% when only the test images were resized to the training average size, and 76.76% when both datasets were standardized to 320x320 before training. (2) New images must go through the same preprocessing.

nnU-Net Results

Metrics	Standardized Results / 1000 epoch (Size:320×320)
Dice	%76.76
IoU	%67.33
Precision	%81.67
Recall	%78.22
Specificity	%98.11
Accuracy	%96.38
Internal Results (BUSBRA)	%91.14

Unet++ / nnUnet Results

Model Architecture	Train Details	BUS-BRA (Internal Validation)	BrEaST (External Test)
UNet++ (ResNet34)	5-fold CV; external test with the best fold (fold 3) checkpoint, TTA + post-processing	%87.40	%70.05
nnUNet (1000 epoch)	Standardized train+test resolution (320×320), 5-fold, 1000 epochs	%91.14	%76.76

Conclusion

Both models demonstrated a decline in performance when transitioning from internal validation to the external BrEaST dataset. UNet++ decreased from 87.40% to 70.05% Dice, while nnU-Net v2 decreased from 91.14% to 76.76% Dice. This consistent drop across both architectures is indicative of the domain shift between the BUS-BRA and BrEaST datasets.

For the nnU-Net v2 pipeline, a significant portion of this performance drop was attributed to image scale inconsistency; standardizing the input resolution (320×320) improved the external Dice to 76.76%.

It is important to note that the internal validation metrics of the two models were obtained under different experimental conditions. UNet++ used a patient-wise cross-validation split, whereas nnU-Net's split was random. Furthermore, the preprocessing and augmentation strategies differed. Therefore, the reported metrics should be viewed as the individual outcomes of two distinct pipelines rather than a direct, controlled comparison. For simulation or robotic inference, the input images must undergo the exact same preprocessing as the training data (min-max normalization and 320×320 bilinear resize).